

BEV Risk Map Analysis Report

Occlusion-Based Emergence Risk Assessment

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Tool: BEV-RiskViz (BEV Risk Visualization Tool)

Purpose: Autonomous driving risk assessment based on occlusion analysis

Scenario Information:

Scenario: Multi-Vehicle Intersection

Risk Calculation Parameters:

weight_trajectory: 0.3

weight_occlusion: 0.3

weight_temporal: 0.2

weight_proximity: 0.2

bev_x_range: (-50.0, 50.0)

bev_y_range: (-50.0, 50.0)

bev_resolution: 0.5

ego_velocity: 10.0

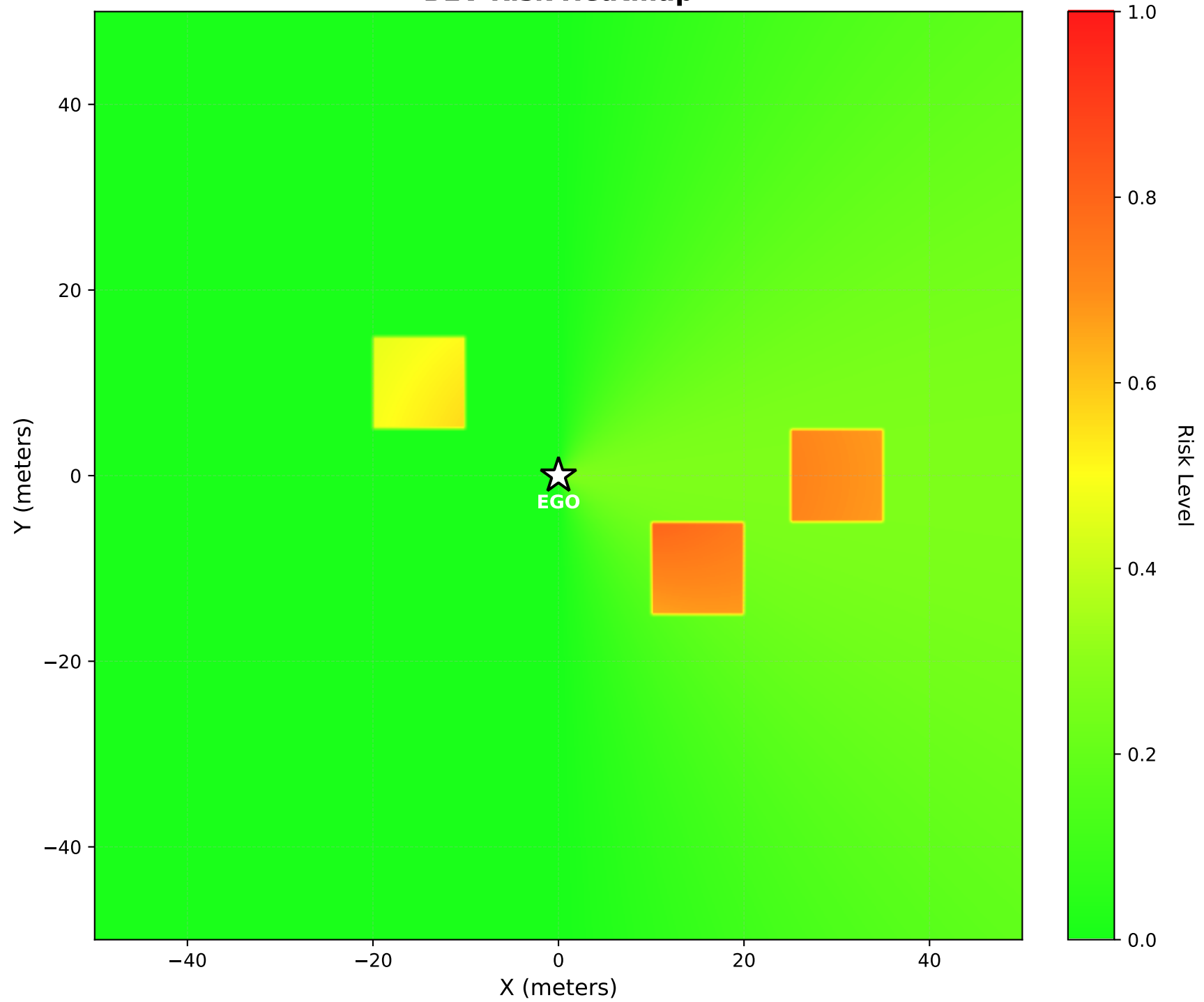
ego_heading: 0.0

max_proximity_distance: 50.0

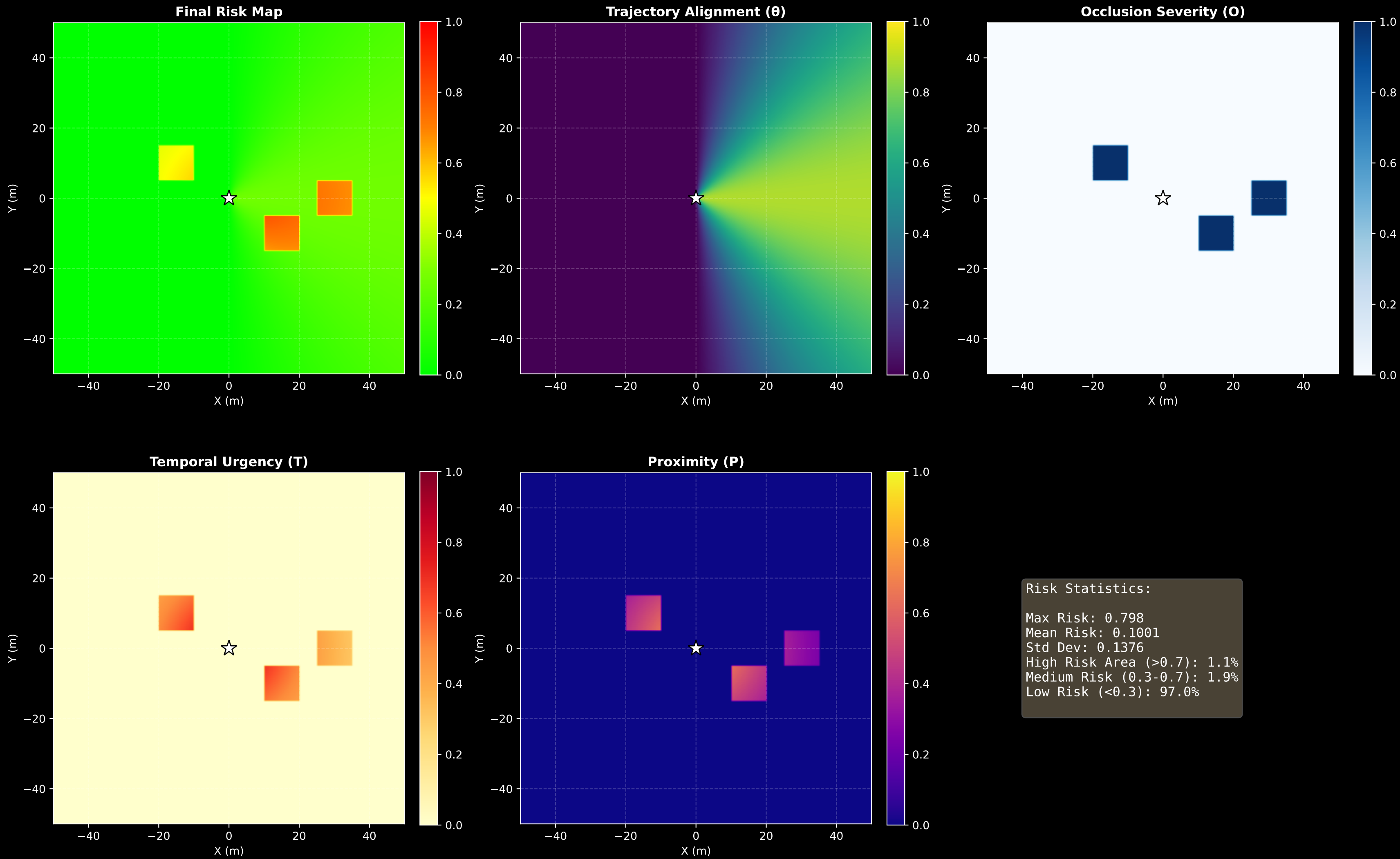
time_horizon: 3.0

occlusion_depth_threshold: 2.0

BEV Risk Heatmap



Risk Factor Analysis



Statistical Analysis

OVERALL RISK STATISTICS

Risk Map Dimensions: 200 × 200 cells
Total Cells: 40,000

Risk Value Statistics:

Maximum Risk: 0.797602
Minimum Risk: 0.000000
Mean Risk: 0.100055
Median Risk: 0.010561
Std Deviation: 0.137647

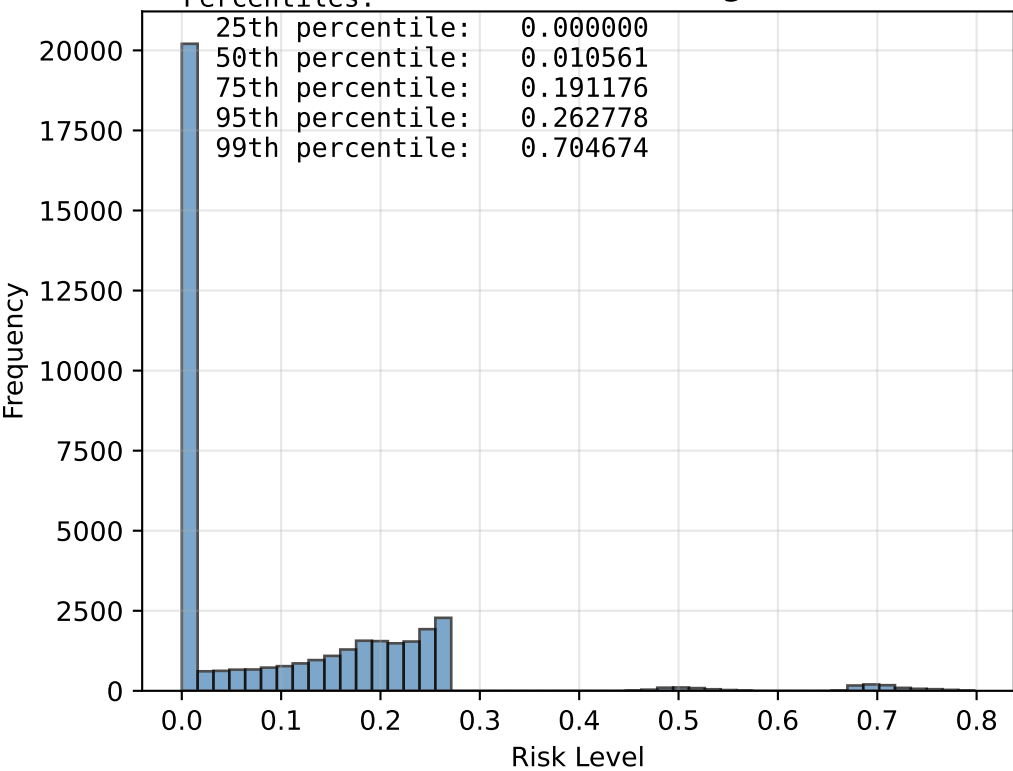
Risk Distribution:

High Risk (>0.7): 454 cells (1.14%)
Medium Risk (0.3-0.7): 746 cells (1.86%)
Low Risk (<0.3): 38,800 cells (97.00%)

Risk Distribution Histogram

Percentiles:

25th percentile: 0.000000
50th percentile: 0.010561
75th percentile: 0.191176
95th percentile: 0.262778
99th percentile: 0.704674



Risk Category Distribution

